

Creating a Creativity Measure

Logan Foster, John Berberian, Sami Adam (Group 7)

Problem Definition

Generative models tend to memorize training data

- Creates privacy concerns
- Conflicts with the broader goal of generating novel data

Generation of Intermediate Digits

- Between 2 and 6
- With some quality-creativity metric
- Compared to latent interpolation of an autoencoder

Challenges

- Generating novel outputs without incentivizing noise
- Defining quality and creativity
- Defining superiority over the baseline

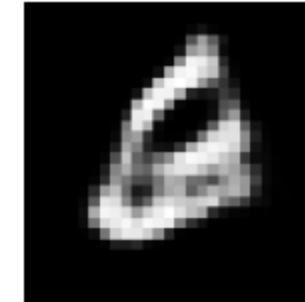
Digit 2



Digit 6



Average Latent



Deep Creativity, Franceschelli et. al.

Creativity Metric is a combination of many factors

- Based on Boden's definition: value+ novelty + surprise
- $DC = a_1V + a_2N + a_3S$ where $\sum(a_i) = 1$

Factor definitions

- **Value:** GAN discriminator scores how real an artifact appears
- **Novelty:** Style classifier measures deviation from known style norms
- **Surprise:** how much the artifact forces weight updates in the generator

Limitations

- Novelty only captures style deviation, not other creative dimensions
- Surprise requires sequential data
- GAN-orientated factors

DeepCreativity: Measuring Creativity with Deep Learning Techniques

Giorgio Franceschelli ^{a,*} and Mirco Musolesi ^b
^aAlma Mater Studiorum Università di Bologna, Italy
E-mail: giorgio.franceschelli@unibo.it
^bUniversity College London, United Kingdom,
The Alan Turing Institute, United Kingdom,
Alma Mater Studiorum Università di Bologna, Italy
E-mail: m.musolesi@ucl.ac.uk

Abstract. Measuring machine creativity is one of the most fascinating challenges in Artificial Intelligence. This paper explores the possibility of using generative learning techniques for automatic assessment of creativity. The proposed solution does not involve human judgement, it is modular and of general applicability. We introduce a new measure, namely *DeepCreativity*, based on Margaret Boden's definition of creativity as composed by *value*, *novelty* and *surprise*. We evaluate our methodology (and related measure) considering a case study, i.e., the generation of 19th century American poetry, showing its effectiveness and expressiveness.

Keywords: Computational Creativity, Deep Learning, Creativity Measure, American Poetry

1. Introduction

Evaluation is a crucial concern in Artificial Intelligence and in science more in general. Measures and metrics are fundamental not only to check the validity of a hypothesis, but also to understand if it is possible to use some given results with confidence as a starting point for future research. An example is Shannon's entropy, which plays a central role as a measure of information, choice and uncertainty [20] and underpins many results in Information Theory [39]. The role of measures is even more crucial in machine learning, where models need to be evaluated during and after training. For example, most of the machine learning algorithms are evaluated using metrics like accuracy.

Excellent progress has been made in benchmark tasks coupled with metrics used to estimate the performance of algorithms [34]: examples include [32] for machine translation or [36] for image generation. At

the same time, there is also a need to derive new metrics for examining the behavior of algorithms in different environments and in relation with society [34]. Among the spectrum of behaviors that could be exhibited by a machine, creativity is certainly one of the most interesting [30]. In fact, we have witnessed the emergence of an entire new field of research, namely Computational Creativity, with a focus on the study of the behaviors exhibited by artificial systems that would be deemed as creative [7,48]. Indeed, one of the key goals of this field is the definition of computational techniques for measuring creativity.

In this paper, we present a novel methodology (and related measure) as a way to assess the creativity of an artifact. In particular, *DeepCreativity* is based on the very famous definition of creativity provided by Margaret Boden: "creativity is the ability to come up with ideas or artifacts that are new, surprising and valuable" [5]. Although it is not the only definition of creativity available (over one hundred of definitions have been proposed over the years [1,42]), it is central in the field of computational creativity. The three aspects that are

^{*}Corresponding author. E-mail: giorgio.franceschelli@unibo.it.

Of Bits and Wows: A Bayesian Theory of Surprise with Applications to Attention, Baldi et. al.

Surprise Definition

- Measures how much a prior belief changes given data
- $S(D, M) = KL(P(M), P(M|D))$
- Always non-negative

Shannon Information Definition

- Measures how rare a data point is
- $I(D) = -\log P(D)$

Surprise vs Shannon Information Tradeoff

- **Many bits, zero wows:** data point that is statistically rare but equally unlikely leaves all beliefs unchanged
- **Few bits, many wows:** data point is not rare but dramatically shifts beliefs

**NIH Public Access**
Author Manuscript
Neural Netw. Author manuscript; available in PMC 2011 June 1.
Published in final edited form as:
Neural Netw. 2010 June ; 23(5): 649–666. doi:10.1016/j.neunet.2009.12.007.

NIH-PA Author Manuscript

NIH-PA Author Manuscript

NIH-PA Author Manuscript

Of Bits and Wows: A Bayesian Theory of Surprise with Applications to Attention

Pierre Baldi^a and Laurent Itti^b
^aDepartment of Computer Science, UCI, Irvine, CA 92697-3435
^bDepartment of Computer Science, USC, Los Angeles, CA 90089-2520

Abstract

The amount of information contained in a piece of data can be measured by the effect this data has on its observer. Fundamentally, this effect is to transform the observer's prior beliefs into posterior beliefs, according to Bayes theorem. Thus the amount of information can be measured in a natural way by the distance (relative entropy) between the prior and posterior distributions of the observer over the available space of hypotheses. This facet of information, termed "surprise", is important in dynamic situations where beliefs change, in particular during learning and adaptation. Surprise can often be computed analytically, for instance in the case of distributions from the exponential family, or it can be numerically approximated. During sequential Bayesian learning, surprise decreases like the inverse of the number of training examples. Theoretical properties of surprise are discussed, in particular how it differs and complements Shannon's definition of information. A computer vision neural network architecture is then presented capable of computing surprise over images and video stimuli. Hypothesizing that surprising data ought to attract natural or artificial attention systems, the output of this architecture is used in a psychophysical experiment to analyze human eye movements in the presence of natural video stimuli. Surprise is found to yield robust performance at predicting human gaze (ROC-like ordinal dominance score ~ 0.7 compared to ~ 0.8 for human inter-observer repeatability, ~ 0.6 for simpler intensity contrast-based predictor, and 0.5 for chance). The resulting theory of surprise is applicable across different spatio-temporal scales, modalities, and levels of abstraction.

Keywords

information; surprise; relative entropy; attention; eye movements

1. Introduction

The concept of information is central to science, technology, and many other human endeavors. While several approaches for quantifying information have been proposed, the most prominent one so far has been Claude Shannon's definition introduced over half a century ago [49,1,39,7,11]. According to this definition, the amount of information contained in a piece of data D is measured by $-\log_2 P(D)$ bits—a rare piece of data with small probability is considered more informative. Although eminently successful for the development of modern telecommunication and computer technologies, Shannon's definition

Publisher's Disclaimer: This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final citable form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

Equations

Value: $V(a, D_v) = D_v(a)$, where D_v is a pre-trained discriminator

- We want our generated data to seem authentic

Novelty $N(a, D_n) = 1 - \frac{\sqrt{\sum_{i=1}^N \left(\frac{1}{N} - y_i\right)^2}}{UB}$ with $UB = \frac{\sqrt{N(N-1)}}{N}$

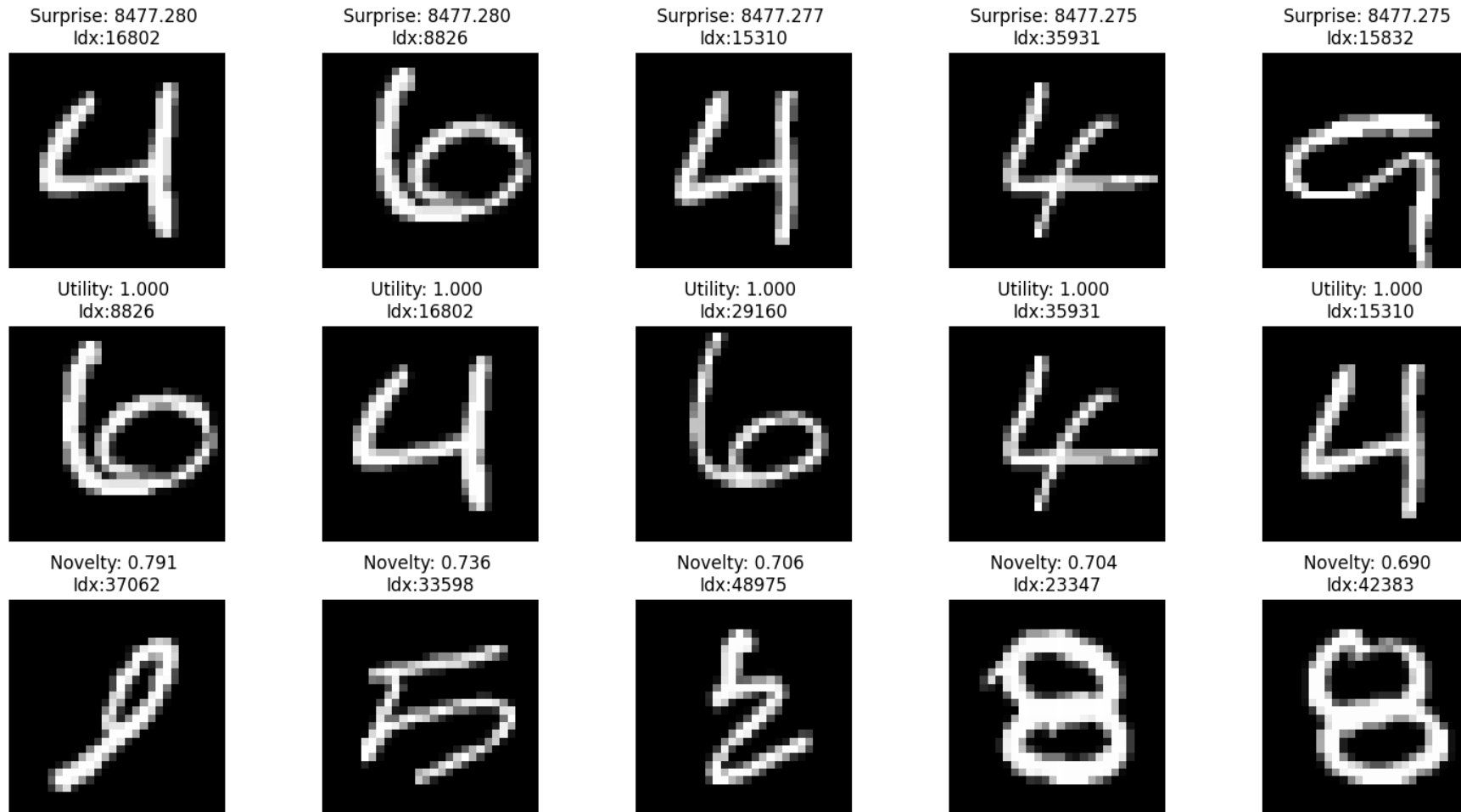
- We want our generated data to deviate from style norms. We estimate this by how our samples don't fit into learned classes.

Surprise $S(a, D_s) = avg_{j,i} \left| \frac{\Delta w_{ji}}{w_{ji}} \right|$

- We want our data to be "surprising." We measure surprise by calculating how much the discriminator's weights would be "corrected" by learning the generated sample was *actually* part of the underlying data distribution.

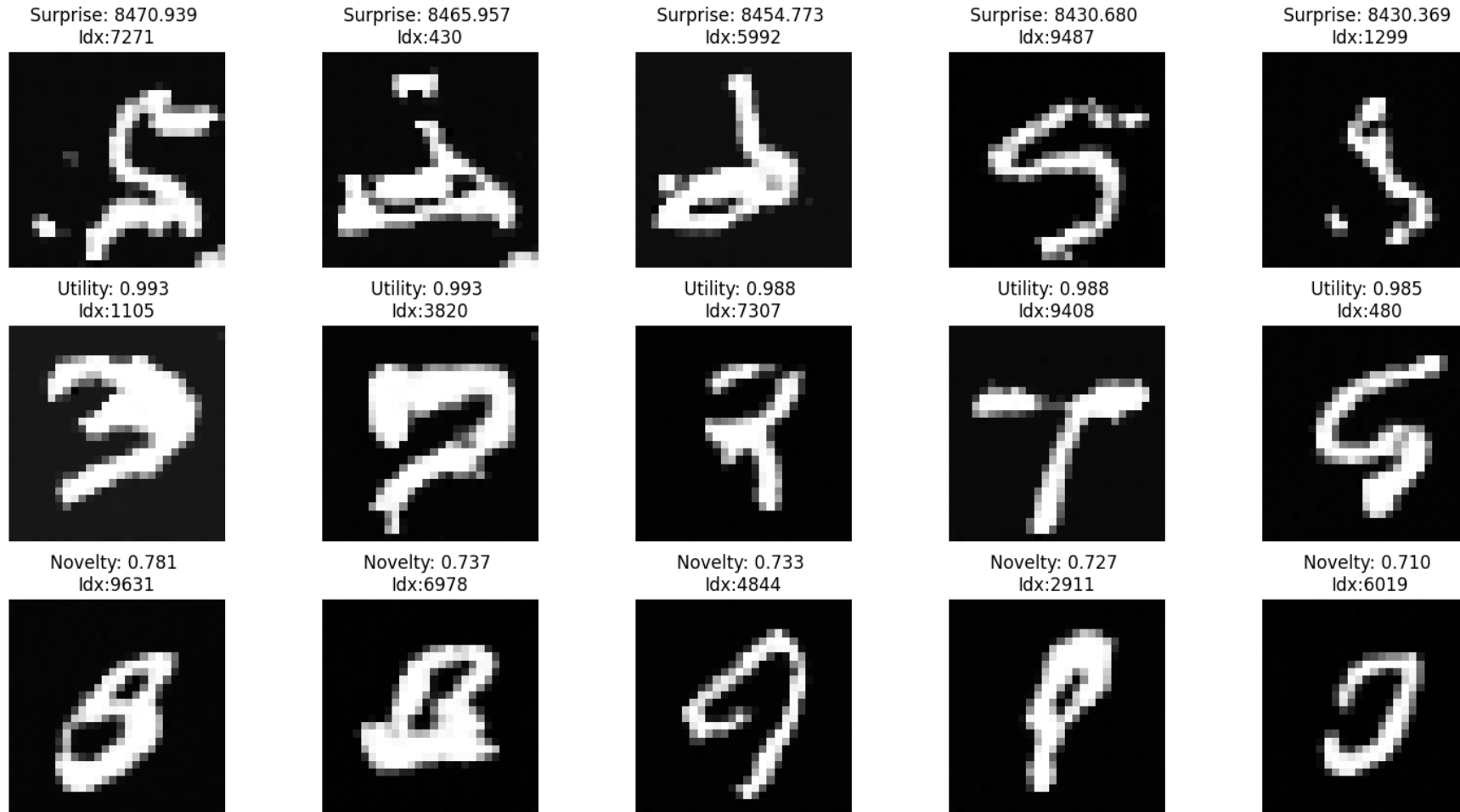
MNIST Images:

Top 5 Images for Each Category (Surprise, Utility, Novelty)



Generated Images:

Top 5 Generated Images for Each Category (Surprise, Utility, Novelty)



Next Steps

Exploring different methods to measure factors

- **Value:** Any classifier trained on the data
- **Novelty:** 2 vs 10 classes
- **Surprise:** Bayesian-style approach (KL or Wasserstein distance)

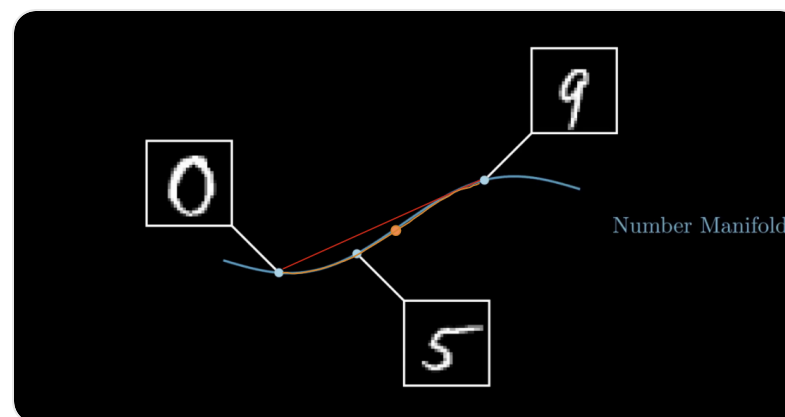
$$N(a, D_n) = 1 - \frac{\sqrt{\sum_{i=1}^N \left(\frac{1}{N} - y_i\right)^2}}{UB} \quad (4)$$

with $UB = \frac{\sqrt{N(N-1)}}{N},$

$$KL(q_\phi(M|D) \| P(M)) \quad \text{vs} \quad KL(P(M) \| q_\phi(M|D))$$

New ways to generate data

- Generate data through interpolations over manifold
(Data Generation without Function Estimation,
Daneshmand et. al.)



Thank you!

Enjoy this GIF of our GAN training progress during Q&A!

